25900325 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSECSC200 - INTRODUCTION TO DATA SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Illustrate the difference between nominal and ordinal data. [U] / [CO2]
2. List any two sources of secondary data. [K] / [CO2]
3. Identify the role of a business analyst in data projects. [U] / [CO1]
4. Examine the scenarios in which you prefer Multiple Linear Regression over Simple Linear Regression and why? [A] / [CO4]
5. List the steps involved in performing EDA. [K] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

6. Calculate mean and standard deviation for the data set: [200,300,400,500,600]. [A] / [CO2]
7. List two advantages of using cross-validation. [K] / [CO4]

8. How are different types of data handled in univariate non-graphical EDA? [An] / [CO3]

9. Illustrate the importance of preprocessing the data. [K] / [CO2]

10. Compare the significance of precision vs Recall in spam filtering. [U] / [CO4]

11. Discuss the importance of detecting outliers during EDA. [U] / [CO3]

12. Compare and contrast the "Modeling" and "Evaluation" stages of the data science life cycle. Explain how the output of one influences the other. [U] / [CO1]

13. Outline the importance of continuous monitoring in model maintenance. [An] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

14. Explain any five commonly used model evaluation matrix in data science. For each, mention whether it is used for regression or classification, and describe what it measures. [U] / [CO4]

15. Break down the role of specialized EDA techniques in the Data Science lifecycle. [U] / [CO3]

16. Explain Data Science and its scope in modern computing. [U] / [CO1]

[15x2 = 30]